

Name: _____



Plotting coordinates generated from a rule

Task A: Finding a rule from a set of coordinates

- 1) For each set of coordinates, fill in the gaps to describe the relationship in words and using algebra:

a) (0,1) (2,3) (9,10)

The y coordinate is equal to the x coordinate _____

$$y = x + 1$$

b) (2,6) (-4,0) (4,8)

The y coordinate is equal to the x coordinate plus 4.

$$y = \underline{\hspace{2cm}}$$

c) (10,5) (4,-1) (0,-5)

The y coordinate is equal to the x coordinate _____

$$y = \underline{\hspace{2cm}}$$

d) (-7,-2) (-3,2) (2,7.2)

The y coordinate is _____

$$y = \underline{\hspace{2cm}}$$

e) (2,4) (3,6) (8,16)

The y coordinate is _____

$$y = \underline{\hspace{2cm}}$$

f) (2,20) (-4,-40) (3,30)

The y coordinate is _____

$$y = \underline{\hspace{2cm}}$$

g) (0,0) (1,5) (-1,-5)

The y coordinate is _____

$$y = \underline{\hspace{2cm}}$$

h) (10,1) (6,0.6) (30,3)

The y coordinate is _____

$$y = \underline{\hspace{2cm}}$$

Task B: Generating coordinates from a rule

- 1) Generate 3 coordinates that follow each rule. Some have the x or y coordinate already filled in.

a) $(3, \quad) (\quad, \quad) (\quad, \quad)$

$$y = x + 20$$

b) $(0, \quad) (\quad, 9) (\quad, \quad)$

$$y = x - 9$$

c) $(0, \quad) (\quad, 300) (\quad, \quad)$

$$y = 100x$$

d) $(0, \quad) (\quad, \quad) (\quad, \quad)$

$$y = 4x$$

- 2) Fill in the table with a different coordinate for the equation that satisfies each condition. The first one has been done for you.

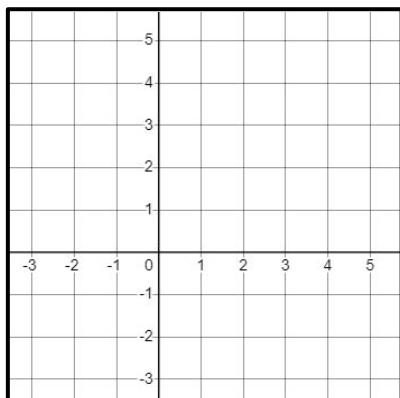
	The y coordinate is positive	The y coordinate is negative	The x coordinate is a non-integer
a) $y = x - 5$	(15,10)	(-5,-10)	(7.5, 2.5)
b) $y = x + 10$			
c) $y = 5x$			
d) $y = -3x$			
e) $y = x - 0.02$			
f) $y = \frac{x}{3}$			



Task C: Plotting coordinates that follow a rule

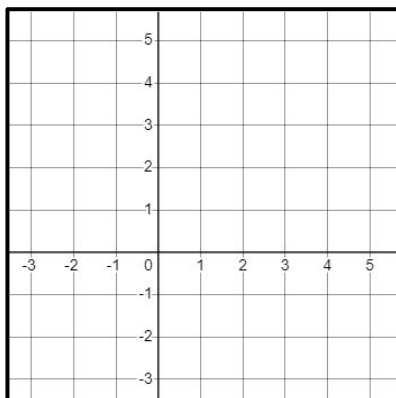
1) Plot all the integer points that satisfy each rule.

a) The x value is 1



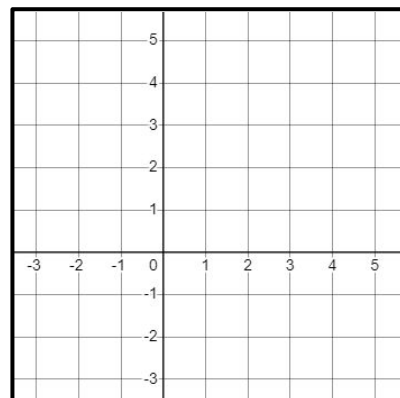
Describe the pattern

b) The y value is one less than the x value



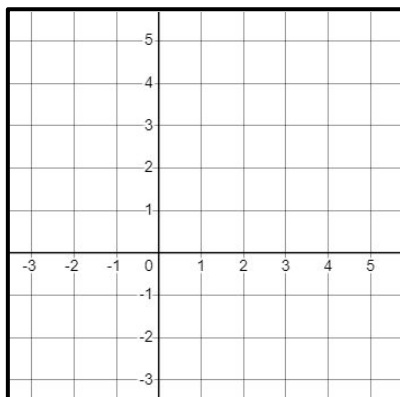
Describe the pattern

c) The y value is double the x value



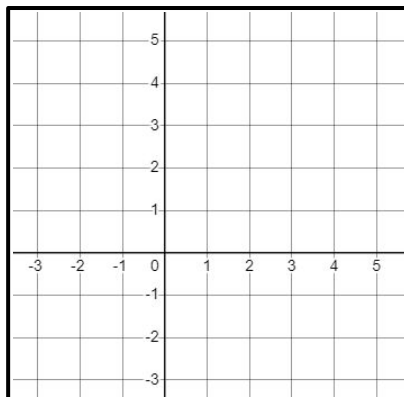
Describe the pattern

d) $y = 0$



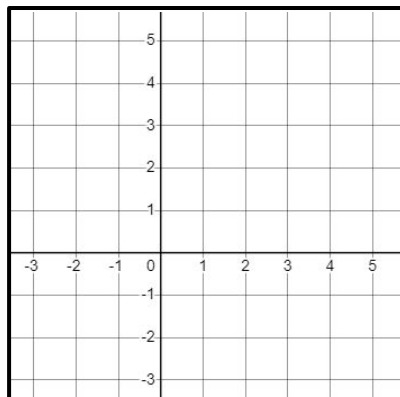
Describe the pattern

e) $y = x - 4$



Describe the pattern

f) $x + y = 5$



Describe the pattern

2) The grid shows seven points labelled A, B, C, D, E, F and G. Complete the table to show which points have coordinates that match the rules below. The first column has been completed for you.

Rule	A	B	C	D	E	F	G
$x = 2$	yes						
$y = -2$	no						
$x = y$	yes						
$y = -x$	no						

